

STILLWATER 5 WNW, OK, USA

WMO: 746430

Lat: 36.1346N

Lon: 97.1082W

Elev: 271

StdP: 98.12

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53927

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-9.0	-16.9	0.9	-10.1	-14.2	1.1	-7.3	7.5	3.0	6.7	2.2	1.9	N/A	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	38.1	23.4	36.1	23.8	34.4	24.0	26.2	33.2	25.6	32.5	25.0	31.9	2.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.8	29.8	23.6	19.1	29.1	23.0	18.4	28.5	82.6	33.2	80.0	32.5	77.5	32.0	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.2	5.4	4.6	-16.9	39.7	4.3	2.6	-20.0	41.6	-22.5	43.1	-24.9	44.5	-28.1	46.4		
			-18.1	27.0	5.3	0.9	-21.9	27.6	-24.9	28.1	-27.9	28.6	-31.7	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	2.6	4.3	10.5	15.1	20.0	25.2	27.4	26.8	22.6	15.6	9.3	3.7
	DBStd	10.03	5.70	6.27	5.96	5.14	4.50	2.93	2.95	3.17	3.86	5.22	5.29	5.32
	HDD10.0	786	237	178	66	13	1	0	0	0	0	11	74	206
	HDD18.3	2146	487	393	250	122	36	1	0	0	9	117	276	455
	CDD10.0	2727	8	18	82	167	311	457	539	522	378	184	52	9
	CDD18.3	1044	0	1	8	26	87	207	281	263	137	32	3	0
	CDH23.3	12056	2	11	84	287	867	2326	3508	3134	1445	357	34	1
	CDH26.7	5749	0	2	14	71	286	1053	1916	1681	622	100	4	0
Wind	WSAvg	1.8	2.0	2.2	2.3	2.4	1.8	1.6	1.3	1.3	1.4	1.6	1.8	1.9
Precipitation	PrecAvg	879	31	36	69	78	124	109	76	78	103	76	58	40
	PrecMax	1179	93	137	196	211	371	268	209	216	315	222	169	130
	PrecMin	500	0	2	0	4	15	21	0	1	1	10	0	2
	PrecStd	171	27	31	47	49	72	60	49	54	66	53	47	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	24.9	28.7	31.5	33.5	37.4	40.8	41.6	36.8	31.6	26.9	21.5
		MCWB	12.3	14.9	17.7	19.0	21.4	23.7	22.7	22.8	21.9	21.8	17.7	13.6
	2%	DB	19.2	21.6	25.6	28.7	31.6	35.0	38.4	38.1	33.7	29.5	23.9	18.5
		MCWB	11.1	13.7	16.1	17.8	21.7	24.0	23.1	23.7	23.0	20.1	15.5	12.6
	5%	DB	16.3	18.6	23.3	26.6	30.0	33.3	36.3	36.1	32.2	27.4	21.5	15.6
		MCWB	10.1	12.6	14.9	17.3	21.4	24.1	23.8	24.0	22.9	18.9	14.6	10.2
	10%	DB	13.0	15.5	20.9	24.7	28.0	31.9	34.5	34.2	30.4	25.1	19.0	13.4
		MCWB	7.6	10.1	13.7	17.2	20.8	23.6	24.4	23.7	22.2	18.2	13.0	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	17.1	19.8	21.4	24.7	26.7	27.1	26.8	25.3	23.4	19.9	17.2
		MCDB	18.3	21.5	26.8	26.9	30.3	33.6	33.6	34.4	32.0	29.2	22.9	20.1
	2%	WB	11.8	15.6	18.1	20.1	23.6	25.8	26.2	25.9	24.6	22.3	17.1	13.7
		MCDB	16.9	20.3	23.8	25.8	29.3	32.4	32.8	33.6	31.2	27.3	21.0	15.4
	5%	WB	9.8	13.7	16.7	19.3	22.6	25.2	25.6	25.2	23.9	21.2	15.6	11.4
		MCDB	15.4	17.7	21.8	24.8	28.1	31.6	32.5	32.6	30.3	25.4	19.7	14.8
	10%	WB	7.8	10.5	14.8	18.2	21.7	24.5	25.1	24.5	23.2	19.7	14.0	9.0
		MCDB	12.3	15.9	19.4	23.3	26.7	30.9	31.9	31.4	29.0	23.3	17.9	12.5
Mean Daily Temperature Range	5% DB	MDBR	13.6	13.1	13.8	13.3	11.8	11.3	12.1	12.5	13.0	13.6	14.0	12.8
		MCDBR	19.9	18.5	16.8	16.0	13.5	13.0	14.8	15.3	14.6	16.0	17.2	17.5
		MCWBR	12.1	11.2	9.5	8.6	6.1	4.7	4.0	4.4	5.3	7.6	9.6	11.7
	5% WB	MCDBR	18.0	16.2	14.3	12.9	11.3	10.9	11.3	11.9	11.6	11.4	13.1	13.9
		MCWBR	11.8	11.0	8.4	7.9	5.9	4.7	4.5	4.6	4.6	6.6	9.1	10.8
Clear-Sky Solar Irradiance	taub	0.297	0.302	0.342	0.382	0.419	0.429	0.434	0.436	0.393	0.348	0.313	0.291	
	taud	2.496	2.493	2.411	2.315	2.283	2.291	2.316	2.253	2.357	2.469	2.504	2.523	
	Ebn at Noon	906	942	927	901	867	855	848	839	861	876	878	891	
	Edh at Noon	81	92	109	127	133	132	128	134	114	93	80	75	
All-Sky Solar Radiation	RadAvg	2.68	3.31	4.32	5.38	5.90	6.74	6.75	6.10	5.07	3.82	2.84	2.26	
	RadStd	0.27	0.49	0.28	0.37	0.60	0.60	0.37	0.37	0.42	0.47	0.36	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page